

properties now. Enter play mode and test it. The Player should be picking up the objects upon contact.

Optimising Memory Usage

Unity calculates all the volumes of all the static colliders in a scene and holds this information in a cache. This saves Unity from recalculating the same information in every frame. Since the Pickup objects are rotating, the volume information is calculated again. And this happens in every frame. So it's not an optimum way of doing things- just uses too much memory. Imagine if you are building an advanced game with hundreds or thousands of rotating static objects which are being recalculated every frame. Now you can image the kind of memory and power usage for an unnecessary task.

Did you note that we mentioned “rotating static objects”? Yep, read that line again. So Unity recalculates the information only for static objects, it does not recache collider volumes for any dynamic objects since they are expected to move. And this is your solution out of this quagmire. You need to indicate to Unity that the Pickup objects are dynamic and not static.

The engine considers any game object with a collider and a rigid body to be dynamic. Any game object with a collider attached but no physics rigid body is expected to be static. The Pickup objects have a box collider but no rigid body so they are static. Select the Prefab asset and add a new component of Rigid Body found under Physics. By adding a rigid body they have become dynamic objects. But this has raised another problem.

If you want to look at the new problem, just hit the play button. All the Pickup objects pass through the Ground and into space. Since we added a rigid body to them, gravitational forces are now acting upon pulling them down and because of the trigger events, they pass through the floor. Apart from gravity, we don't want other forces to also act upon them which may create possible problems in the future. So in the Rigid Body component, select the 'Is Kinematic' option. So it becomes a Kinematic rigid body which does not react to physical forces. Now the option of “Use Gravity” selected or not is moot since gravity is also a physical force and will not be applied here.

If you test now, everything works smoothly and is perfectly optimised. **d**